ÉTIENNE ZONON

Senior Data Scientist and Statistician-Economist

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PROFESSIONAL PROFILE

Senior Data Scientist and Statistician-Economist with over 6 years of experience applying advanced econometrics, machine learning, and emerging AI technologies to support evidence-based decision-making. Proven track record in designing predictive models, macrofinancial analysis, and time-series forecasting within financial and telecom sectors. Expert at leading cross-functional analytical initiatives and translating complex quantitative research into strategic insights for technical and non-technical stakeholders. Fully bilingual (English/French), blending strong economic modeling with production-grade engineering in cloud-native AI systems.

CORE QUALIFICATIONS

• **Quantitative Analysis & Modeling:** Econometrics, Statistical Modeling, Time-Series Forecasting (ARIMA, Prophet), Predictive Modeling, Customer Segmentation, Machine Learning (Random Forest, SVM, PCA)
• **Emerging AI & GenAI:** Generative AI, Large Language Models (LLMs), Agentic AI Architectures, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP)
• **Tools & Cloud Platforms:** Python, PySpark, R, SQL (Oracle T-SQL, PostgreSQL), Databricks, Microsoft Fabric, Azure, Git/GitHub
• **Business & Governance:** Reproducible Research, Model Governance, Responsible AI, Stakeholder Engagement, Cross-Functional Collaboration, Agile/Scrum

PROFESSIONAL EXPERIENCE

AI Developer & Data Engineer — *4ChangeHub, Canada (Remote)* Aug. 2025 – Present

- Design and industrialize end-to-end data products on Microsoft Fabric (Lakehouse, Data Factory, Warehouse), establishing reliable, reproducible foundations for reporting and strategic analysis.
- Implement scalable NLP pipelines and enterprise RAG architectures using LangChain and PostgreSQL to automate complex information extraction and expand organizational data access.
- Transform, clean, and validate massive datasets using PySpark and Python, optimizing data ingestion and enforcing strict quality and anomaly detection protocols.
- Partner closely with multidisciplinary business stakeholders and technology teams to align analytical solutions with organizational performance measurement goals.

Technologies used: Microsoft Fabric, Python, PySpark, PostgreSQL, LangChain, FastAPI, Streamlit, Power BI, Scikit-learn, SQL

Data Scientist — *Orange Liberia, Remote* Oct. 2024 – Apr. 2025

- Spearheaded quantitative analysis of large-scale telecom datasets and key financial performance metrics (ARPU, AUPU, AUPA) to identify structural growth levers and improve operational efficiency by 15%.
- Engineered and deployed statistical anomaly-detection models, decreasing processing errors by 20% while building sustainable analytical tracking capabilities.
- Translated complex data findings into actionable insights for senior leadership through interactive Power BI systems, bridging the gap between technical methodologies and executive strategy.

Technologies used: Python, SQL, Scikit-learn, Power BI, Statistical Analysis, Anomaly Detection, Telecom KPIs

Data Scientist – Marketing — *Orange Côte d'Ivoire, Abidjan, Côte d'Ivoire* Jan. 2023 – Sept. 2024

Conducted advanced customer behavior modeling, propensity scoring, and likelihood-to-buy forecasting in a distributed
Technologies used: Spark/PySpark, Python, Scikit-learn, Oracle SQL (T-SQL), Power BI, PCA, Scrum, Trello

- **Data Scientist – Technical** Big Data cluster to maximize client retention and support cross-selling.
- Executed complex dimensionality reduction (PCA) and advanced statistical segmentation to power granular recommendation engines and optimize marketing resource allocation.
- Industrialized SQL indicators (Oracle T-SQL), automated core reporting workflows, and cut campaign overhead tracking time by 40% using Agile/Scrum frameworks.
- Analyzed model feature importance to translate complex statistical metrics into clear, actionable insights for marketing and finance teams.

Services — *Orange Côte d'Ivoire, Abidjan, Côte d'Ivoire* Dec. 2019 – Dec. 2022

- Developed advanced statistical forecasting models for complex network performance time series leveraging R, ARIMA, and Facebook Prophet to support infrastructure planning.
- Programmed automated web scraping scripts to extract and process logs from the Huawei U2000 network platform, minimizing manual data engineering overhead.
- Designed and launched an interactive R Flexdashboard application to consolidate technical analyses, reducing manual reporting requirements by approximately one week per month.

Technologies used: R, Python, Facebook Prophet, ARIMA, Scikit-learn (Random Forest, SVM), R Flexdashboard, Web Scraping

COMPLEMENTARY PROJECTS & RESEARCH DEMONSTRATIONS

Macrofinancial Modeling Framework (BRVM Macrofinance Scout) — Built an automated text-mining and macroeconomic modeling framework for macro-desk research. Developed a text-processing engine using LLMs (via Groq/OpenAI APIs) to ingest financial feeds, transforming text into structured sentiment metrics (+0.44 net sentiment). Implemented and compared multiple predictive models to forecast market movements: Multiple Linear Regression (achieving an R^2 of 0.68 based on Money Supply Growth and Bank Capital Strength), Logistic Regression (68% short-term up-probability), and Random Forest Classification for volatility regime mapping (73% confidence).

Technologies used: Python, OpenAI/Groq API, NLP Topic Modeling, Scikit-learn, Linear/Logistic Regression, Random Forest

Agentic AI Telegram Assistant & Document Processor — Engineered a fully localized, open-source RAG conversational assistant utilizing Ollama (local LLMs) and Qdrant vector databases, tracking performance metrics via MLflow. Developed an AI-powered document extraction pipeline using Qwen2.5-Coder to systematically transform unstructured scans into validated JSON/Excel data formats with automated discrepancy verification.

Technologies used: Ollama, Qdrant, Embeddings, LangChain, MLflow, Qwen2.5-Coder, Python, Git/GitHub

EDUCATION & CONTINUOUS LEARNING

Software Engineering Technology - Artificial Intelligence — *Centennial College, Online - Continuous Learning* Jan. 2026 – Now

Focus: Cloud AI solutions, distributed data streaming, scalable software architectures, and automated discrepancy validation.

Master's Degree in Data Science & Big Data — *INP-HB, Côte d'Ivoire* Sept. 2018 – Dec. 2019

WES Equivalency Available.

Engineer - Statistician-Economist — *ISSEA, Cameroon* Sept. 2015 – July 2018

Advanced quantitative methods, macroeconomics, structural econometric modeling, and economic forecasting.

CERTIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate (DP-700) & Databricks Data Engineer Associate (DP-750)
- Microsoft Certified: SQL AI Developer (DP-800) & AI Fundamentals (AI-900)
- Expert Credentials: Advanced Data Science with IBM & Generative AI with Large Language Models (DeepLearning.AI)
- Frameworks & Agility: Google Project Management & Agile / Scrum Master (LearnQuest)

LANGUAGES

- **English:** Professional working proficiency (Experienced in English-essential professional environments)
- **French:** Native proficiency